



Uniform indirect boundary observability for a spatial discretization of strongly coupled wave equations

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ABSTRACT

This paper investigates the indirect boundary observability properties of one-dimensional strongly coupled wave equations in an approximated setting. Classical numerical discretization methods, such as finite differences and finite elements, typically fail to maintain uniform observability inequalities when applied to wave systems. This failure is primarily attributed to the emergence of high-frequency numerical solutions. The present work demonstrates a different approach through the implementation of these discretization schemes on a carefully designed non-uniform mesh. This study successfully establishes uniform observability inequalities for the coupled system. This methodology effectively recovers the system's total energy through boundary observations, overcoming the well-documented limitations of traditional numerical approaches in wave equation systems.

1. Introduction

In this study, we investigate the observability property of space semi-discretizations for the following system of coupled wave equations:

$$\begin{cases} \phi_{tt} - \phi_{xx} + \alpha\psi_t = 0, & (x, t) \in (0, 1) \times (0, T), \\ \psi_{tt} - \psi_{xx} - \alpha\phi_t = 0, & (x, t) \in (0, 1) \times (0, T), \\ \phi(0, t) = \phi(1, t) = 0, & t \in (0, T), \\ \psi(0, t) = \psi(1, t) = 0, & t \in (0, T), \\ \phi(x, T) = \phi^0(x), \quad \phi_t(x, T) = \phi^1(x), & x \in (0, 1), \\ \psi(x, T) = \psi^0(x), \quad \psi_t(x, T) = \psi^1(x), & x \in (0, 1), \end{cases} \quad (1.1)$$

where $\alpha \in \mathbb{R}^*$ is coupling parameter. The coupling terms ϕ_t and ψ_t come from classical models of stacked elastic bodies. These models describe interaction through inertial or visco-inertial effects. Energy exchange depends on velocities, not on direct shear strain. This type of coupling is justified when the interface behaves like a thin fluid or viscoelastic layer. Such layers allow slip between bodies. It also applies to piezoelectric materials, where electromechanical effects depend on displacement rates. Velocity-based coupling appears in real applications. For example, in the Rao–Nakra sandwich beam model.¹ Unlike standard shear-type couplings of the form $(\phi_t - \psi_t)$, which resist relative velocity through shear stress,² the velocity-only form allows slip. It models interfaces that transmit forces proportional to velocity. This makes it suitable for fluid-like or electromechanically active layers.

Since the analysis of observability at the discrete level is generally based on the continuous framework, it is useful to shed light on the observability results for the system in (1.1).

In Ref. 3, various types of observability inequalities were examined based on the algebraic properties of the coupling parameter α . When α is positive and takes small values, a specialized approach utilizing multiplier techniques demonstrated that the system (1.1) is observable in the energy space $\mathcal{H} := (H_0^1(0, 1) \times L^2(0, 1))^2$. More precisely, the following inequality holds:

$$\mathcal{E}(0) \leq C(\alpha, T) \int_0^T |\psi_x(1, t)|^2 dt, \quad (1.2)$$

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for any solution $(\varphi, \varphi_t, \psi, \psi_t)$ of the backward adjoint system associated with Eq. (1.1), given final data $(\varphi^0, \varphi^1, \psi^0, \psi^1)$ in the energy space $\mathcal{H}$. The energy associated with the system is given by:

$$\mathcal{E}(t) = \int_0^1 |\varphi_t|^2 dx + \int_0^1 |\varphi_x|^2 dx + \int_0^1 |\psi_t|^2 dx + \int_0^1 |\psi_x|^2 dx. \quad (1.3)$$

Due to the hyperbolic nature of wave systems, the observability estimate (1.2) holds only for observation times T exceeding a critical threshold T_0 .

In the one-dimensional single wave equation, the optimal observability time is $T_{obs} = 2$, according to the Geometric Control Condition (GCC),^{4,5} which depends on the wave speed and the length of the domain. In our setting, the coupled system (1.1) involves modes propagating at different effective speeds, which naturally depend on the coupling parameter α . Therefore, in view of the GCC, the minimal observability time T_0 becomes a function of α , and it is generally different from T_{obs} . In Ref. 3, the authors compute T_0 based on algebraic properties of the coupling parameter α , and various observability inequalities are also discussed in that context.

In the context of coupled systems, observability estimates such as (1.2) are particularly practical, as the observation involves only one component. This type of observability is referred to as indirect observability and has also been studied in the context of weakly coupled systems.⁶ Indirect observability is more advantageous compared to observing all components simultaneously, a situation referred to as “complete observability”, which has been explored in Refs. 7, 8.

This work analyzes a discrete version of the observability inequality (1.2) using finite differences and finite element techniques on a non-uniform mesh. The approach aims to establish uniformity of the observability estimate with respect to mesh size, which is critical for constructing uniformly bounded discrete controls for the discrete controlled system. This implication is grounded in duality principles within a finite-dimensional framework.⁹ The interest of numerical control problems lies in wave equations, specifically when implementing the Hilbert uniqueness method for wave control, using finite classical differences and finite element methods.¹⁰ The authors discovered a significant numerical challenge: the numerical control deviates from the continuous control as mesh refinement increases, particularly when initial data contains high-frequency components. Theoretical investigations in one and two-dimensional cases examined this phenomenon from an observability perspective, revealing a blow-up of discrete observability as mesh size approaches zero,^{11,12} accompanied by divergence of corresponding discrete controls.⁹ This undesirable result is also observed in Ref. 13 for the coupled system (1.1), discretized using finite differences and finite elements on a uniform mesh. To address the challenges posed by spurious high-frequency components in numerical observability and establish uniform observability inequalities, researchers have developed several methodological approaches. Fourier filtering techniques¹¹ constrain solutions within a truncated Fourier subspace, though implementation remains problematic due to implicit confinement constraints. The multigrid methodology presents a more explicit spatial filtering framework.¹⁴ Alternative methodologies encompass the utilization of high-regularity initial conditions to guarantee uniform boundary observability,¹⁵ and Tikhonov regularization incorporating interior mesh point observations.¹⁰ The mixed finite element formulation constitutes a promising discretization paradigm, offering an effective approach to mitigate high-frequency numerical artifacts and enhance discretization stability.¹⁶ Recent extensions of these remedies to coupled systems have been demonstrated in contemporary literature.^{17–20}

Introducing a suitable vanishing numerical viscosity into finite difference or finite element discretizations on a uniform mesh is a well-established analytical strategy to suppress high-frequency spurious modes that typically compromise uniform properties. This methodology has proven effective in the context of the one-dimensional wave equation, where it ensures uniform exponential decay of the discrete energy.²¹ Uniform boundary controllability has also been established using this approach.²² Its effectiveness has recently been extended to weakly coupled wave systems for indirect boundary controllability.²³

A finite difference based scheme relying on inertia averaging has recently been developed to preserve uniform observability on uniform meshes. For the one-dimensional wave equation, an order-reduction finite difference scheme was introduced in Ref. 24, achieving uniform exponential decay of the discrete energy without requiring numerical viscosity. Similar techniques have been successfully applied to the beam equation, ensuring uniform observability even under fully clamped boundary conditions,²⁵ and uniform exponential stability under non-located boundary feedbacks.²⁶ This method has been further extended to weakly coupled systems such as multilayered smart structures.^{2,27} In the context of strongly coupled systems, applications to piezoelectric beams and thermal effects were considered in Refs. 28, 29, where uniform stabilization was also retained. These contributions highlight the effectiveness of inertia-averaging schemes as robust alternatives to classical discretizations, particularly for boundary-controlled problems.

This study implements a non-uniform mesh framework, constructed via concave functions, to achieve uniform observability in coupled wave systems. Building upon the methodology established in Ref. 30, which demonstrated uniform observability for single wave equations, we analyze the coupling parameter's influence on system behavior. The approach addresses the fundamental challenge of high-frequency solutions propagating at diminishing velocities as mesh size decreases. By implementing strategic mesh refinement near observation boundaries, the method effectively manages energy distribution, ensuring high-frequency components remain negligible compared to boundary-concentrated energy. This refinement strategy facilitates the establishment of uniform observability estimates within the coupled system framework.

The structure of this paper is as follows: In Section 2, we introduce the space semi-discrete formulation of the system (1.1) on a non-uniform mesh. Section 3 presents the principal findings of our study. Sections 4 and 5 are devoted to proving these results for the finite difference and finite element cases, respectively. Lastly, Section 6 concludes the paper with remarks and future perspectives.

2. Discretizations

In this section, we consider the space semi-discretization of the system (1.1) using finite difference and finite element methods on a non-uniform mesh. Let $g : [0, 1] \rightarrow [0, 1]$ be a diffeomorphic function. The non-structured mesh of the interval $[0, 1]$ is generated by the function g as follows:

$$\mathcal{M}^{h,g} = \{x_j = g(jh), j = 0, \dots, N+1\}, \quad (2.1)$$

where $h = 1/(N+1)$, $N \in \mathbb{R}^*$ is the mesh size parameter of the uniform mesh. We define the following quantities required for the discretizations:

$$h_{j+1/2} = x_{j+1} - x_j, \quad j = 0, \dots, N, \quad (2.2)$$

and

$$h_j = \frac{h_{j+1/2} + h_{j-1/2}}{2}, \quad j = 1, \dots, N. \quad (2.3)$$

Here, $h_{j+1/2}$ represents the distance between adjacent points in the mesh, and h_j is the average of the distances to the neighboring points.

2.1. Finite difference-based scheme

The finite difference discretization of the continuous system (1.1) is straightforwardly obtained as follows:

$$\begin{cases} h_j \phi_j'' - \left(\frac{\phi_{j+1} - \phi_j}{h_{j+1/2}} - \frac{\phi_j - \phi_{j-1}}{h_{j-1/2}} \right) + \alpha h_j \psi_j' = 0, & j = 1, \dots, N, \quad t \in (0, T), \\ h_j \psi_j'' - \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} - \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) - \alpha h_j \phi_j' = 0, & j = 1, \dots, N, \quad t \in (0, T), \\ \phi_0 = 0, \quad \phi_{N+1} = 0, \quad \psi_0 = 0, \quad \psi_{N+1} = 0, & t \in (0, T) \\ \phi_j(0) = \phi_j^0, \quad \phi_j'(0) = \phi_j^1, \quad \psi_j(0) = \psi_j^0, \quad \psi_j'(0) = \psi_j^1. \end{cases} \quad (2.4)$$

Here, $\phi_j(t) \approx \phi(x_j, t)$ and $\psi_j(t) \approx \psi(x_j, t)$. Given any initial states $\phi_h^k = (\phi_1^k, \dots, \phi_N^k)$ and $\psi_h^k = (\psi_1^k, \dots, \psi_N^k)$, for $k = 0, 1$ in $(\mathbb{R}^N \times \mathbb{R}^N)^2$, we define $(\phi_h, \psi_h) = [\phi_1, \dots, \phi_N, \psi_1, \dots, \psi_N]$, as the unique solution of the discretized system (2.4). The discrete energy of this solution, $\mathcal{E}_h(t)$, is given by:

$$\begin{aligned} \mathcal{E}_h(t) = & \frac{1}{2} \sum_{j=1}^N h_j |\phi_j'|^2 + \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\phi_{j+1} - \phi_j}{h_{j+1/2}} \right|^2 \\ & + \frac{1}{2} \sum_{j=1}^N h_j |\psi_j'|^2 + \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2. \end{aligned} \quad (2.5)$$

2.2. Finite element-based scheme

The variational formulation of the system (1.1) aims to find functions $\phi(t)$ and $\psi(t)$ in the Hilbert space $H_0^1(0, 1)$ such that:

$$\begin{cases} \frac{d^2}{dt^2} \int_0^1 \phi u dx + \int_0^1 \phi_x u_x dx + \alpha \frac{d}{dt} \int_0^1 \psi u dx = 0, & \forall u \in H_0^1(0, 1), \\ \frac{d^2}{dt^2} \int_0^1 \psi u dx + \int_0^1 \psi_x u_x dx - \alpha \frac{d}{dt} \int_0^1 \phi u dx = 0, & \forall u \in H_0^1(0, 1). \end{cases} \quad (2.6)$$

For discretizing (2.6), we use the classical functions u_i , $i = 1, \dots, N$, as defined in¹¹:

$$u_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_j - x_{i-1}}, & x \in [x_{i-1}, x_i], \\ \frac{x_{i+1} - x}{x_{i+1} - x_j}, & x \in [x_j, x_{i+1}], \\ 0, & \text{otherwise.} \end{cases} \quad (2.7)$$

Using these basis functions, the solutions are approximated as

$$\phi_h(t) = \sum_{i=1}^N \phi_i(t) u_i, \quad \psi_h(t) = \sum_{i=1}^N \psi_i(t) u_i. \quad (2.8)$$

Substitution of the functions from (2.8) into the system (2.6) yields the following reduced form:

$$\begin{cases} \frac{d^2}{dt^2} \int_0^1 \phi_h u_j dx + \int_0^1 \phi_{hx} u_{jx} dx + \alpha \frac{d}{dt} \int_0^1 \psi_h u_j dx = 0, & \forall j = 1, \dots, N, \\ \frac{d^2}{dt^2} \int_0^1 \psi_h u_j dx + \int_0^1 \psi_{hx} u_{jx} dx - \alpha \frac{d}{dt} \int_0^1 \phi_h u_j dx = 0, & \forall j = 1, \dots, N. \end{cases} \quad (2.9)$$

The evaluation of the following integrals, in combination with (2.9),

$$\int_0^1 u_i u_j dx = \begin{cases} \frac{x_j - x_{j-1}}{6}, & |i - j| = 1, \\ \frac{x_{j+1} - x_{j-1}}{3}, & i = j, \\ 0, & |i - j| \geq 2, \end{cases} \quad \int_0^1 u_{ix} u_{jx} dx = \begin{cases} \frac{2}{h}, & |i - j| = 1, \\ -\frac{1}{h}, & i = j, \\ 0, & |i - j| \geq 2, \end{cases} \quad (2.10)$$

yields the finite element discretization:

$$\begin{cases} \frac{h_{j+1/2}}{6} \phi_{j+1}'' + \frac{2h_j}{3} \phi_j'' + \frac{h_{j-1/2}}{6} \phi_{j-1}'' - \left(\frac{\phi_{j+1} - \phi_j}{h_{j+1/2}} - \frac{\phi_j - \phi_{j-1}}{h_{j-1/2}} \right) \\ \quad + \alpha \left(\frac{h_{j-1/2}}{6} \psi_{j+1}' + \frac{2h_j}{3} \psi_j' + \frac{h_{j-1/2}}{6} \psi_{j-1}' \right) = 0, & j = 1, \dots, N \\ \frac{h_{j+1/2}}{6} \psi_{j+1}'' + \frac{2h_j}{3} \psi_j'' + \frac{h_{j-1/2}}{6} \psi_{j-1}'' - \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} - \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) \\ \quad - \alpha \left(\frac{h_{j+1/2}}{6} \phi_{j+1}' + \frac{2h_j}{3} \phi_j' + \frac{h_{j-1/2}}{6} \phi_{j-1}' \right) = 0, & j = 1, \dots, N \\ \phi_0 = 0, \quad \phi_{N+1} = 0, \quad \psi_0 = 0, \quad \psi_{N+1} = 0, \\ \phi_j(0) = \phi_j^0, \quad \phi_j'(0) = \phi_j^1, \quad \psi_j(0) = \psi_j^0, \quad \psi_j'(0) = \psi_j^1. \end{cases} \quad (2.11)$$

For any initial states $\varphi_h^k = (\varphi_1^k, \dots, \varphi_N^k)$ and $\psi_h^k = (\psi_1^k, \dots, \psi_N^k)$, for $k = 0, 1$ in $\mathbb{R}^N$, we define $(\varphi_h, \psi_h) = [\varphi_1, \dots, \varphi_N, \psi_1, \dots, \psi_N]$, as the unique solution of the discretized system (2.11). The discrete energy of this solution, $\mathcal{E}_h(t)$, is given by:

$$\begin{aligned} \mathcal{E}_h(t) = & \frac{1}{12} \sum_{j=0}^N h_{j+1/2} |\varphi'_j + \varphi'_{j+1}|^2 + \frac{1}{6} \sum_{j=1}^N h_j |\varphi'_j|^2 + \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 \\ & + \frac{1}{12} \sum_{j=0}^N h_{j+1/2} |\psi'_j + \psi'_{j+1}|^2 + \frac{1}{6} \sum_{j=1}^N h_j |\psi'_j|^2 + \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2. \end{aligned} \quad (2.12)$$

For both finite difference and finite element methods, the solutions (φ_h, ψ_h) and energies $\mathcal{E}_h$ must reflect their dependence on the function g in the notation. However, for simplicity, we omit this dependence from the notation.

3. Main results

In this section, we present the principal contributions of this work. These include two main results: the discrete formulation of the observability inequality (1.2) within the framework of the finite difference scheme (2.4) and the finite element scheme (2.11).

Theorem 3.1. *There exist constants $T_0 > 0$ and $\hat{\alpha} > 0$ for which the following holds: For any $T > T_0$ and $0 < \alpha < \hat{\alpha}$ there exists a smooth diffeomorphism g on $[0, 1]$ and positive constant $C(\alpha, T, g)$ such that the following estimate is satisfied for any initial state $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ of the system (2.4):*

$$\mathcal{E}_h(0) \leq C(\alpha, T, g) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt, \quad (3.1)$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.4).

In the context of a single wave, it has been shown that observability inequality cannot be established uniformly with respect to h when using classical finite difference and finite element methods.¹¹ This issue arises because solutions with high frequencies propagate at velocities that approach zero as $h \rightarrow 0$, preventing them from reaching the observation boundary ($x = 1$). Particularly with a uniform mesh, the total energy becomes increasingly concentrated away from the observation boundary as the mesh is refined. Consequently, the observability estimate fails to be uniform.

A similar non-uniform behavior has been observed for the coupled system (1.1) when applying finite difference and finite element schemes on uniform meshes.¹³ In particular, the observability constant associated with estimate (3.1), for the choice $g(x) = x$ exhibits, blow-up as $h \rightarrow 0$.

To address this problem, the natural solution is to refine the mesh near the observation boundary, ensuring that the energy is dominated by its value in the region close to the boundary where the solution vanishes. By designing the mesh in such a way, the total energy for high-frequency solutions becomes negligible compared to the energy concentrated at the observation boundary, allowing for a uniform observability estimate. This approach is effectively implemented by the authors of Ref. 30, who use a concave diffeomorphism g to refine the mesh near the observation boundary.

In Theorem 3.1, the uniformity property is achieved by employing a concave diffeomorphism function with similar properties in Ref. 30, as detailed in Section 4 of this paper. The same ideas are adopted to achieve a uniform observability inequality for system (2.11), as stated in the following result.

Theorem 3.2. *There exist constants $T_0 > 0$ and $\hat{\alpha} > 0$ for which the following holds: For any $T > T_0$ and $0 < \alpha < \hat{\alpha}$ there exists a smooth diffeomorphism g on $[0, 1]$ and positive constant $C(\alpha, T, g)$ such that the following estimate is satisfied for any initial state $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^n \times \mathbb{R}^N)^2$ of the system (2.11):*

$$\mathcal{E}_h(0) \leq C(\alpha, T, g) \left[\int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + \frac{1}{6} \int_0^T |\psi'_N|^2 dt \right], \quad (3.2)$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.11).

The observability inequality (3.2) includes the term ψ'_N in the observation integral, a term that does not appear in its continuous counterpart (1.2). The presence of this additional term is well-known in finite element-based schemes,^{11,16,30} and it arises due to the mass matrix acting on the second-order time derivative. However, this term can be rewritten as

$$\frac{1}{6} \int_0^T |\psi'_N|^2 dt = \frac{h^2}{6} \int_0^T \left| \frac{\psi'_N}{h} \right|^2 dt, \quad (3.3)$$

and, intuitively, under suitable regularity assumptions on the final condition, this contribution tends to zero as $h \rightarrow 0$. This behavior has been rigorously established in Ref. 13 for the mixed finite element method on a uniform mesh. By duality, in the case of the single wave equation, despite the presence of this additional term, one can construct discrete controls that converge, provided high frequency filtering conditions are imposed.^{9,16}

The problem of observability for wave systems requires a minimal observation time T . For a single wave equation, the minimal observation time in the one-dimensional case is $T_{obs} = 2$, a condition that originates from the physical properties of wave propagation. A similar principle applies to coupled wave systems. Moreover, the set of coupling parameters for which the system is naturally observable also has a physical origin. The theoretical identification of these two properties can depend on the approach used. In Ref. 3, using the multiplier technique, the observability was established for time intervals T exceeding a threshold $T_0 > 0$ and for coupling parameter within $(0, \hat{\alpha})$. For practical implementation, the non-structured mesh based on function g must be designed to accommodate any given values of T and α . In Theorem 1 of Ref. 30, the author establishes uniform boundary observability for the single wave equation through a two-step analysis. First, uniform observability is demonstrated for $T > T_g$. Subsequently, by employing a specific choice of g as defined in Eq. (3.4), uniform observability is recovered for $T > T_{obs}$:

$$g_\theta = \sqrt{(2\theta + 1)x + \theta^2} - \theta \quad \theta \in \left(0, \frac{T-2}{2}\right). \quad (3.4)$$

Since we are dealing with wave systems, we apply the methodology from Ref. 30 to explore its applicability in our context. For simplicity, we omit the first step and directly employ the diffeomorphisms defined in Eq. (3.4) to establish the uniform estimates given in Eqs. (3.1) and (3.2). While in the single wave context, θ can be freely chosen for any time T , ($\theta \in]0, (T-2)/2[$) this flexibility is lost in our setting due to the coupling parameter and other considerations, which will be discussed in Remark 4.8.

4. Proof of Theorem 3.1

This section focuses on establishing the uniform estimate presented (3.1). While the necessary strict concavity property of function g is well understood, additional smoothness conditions are required due to technical considerations. Following³⁰ the smoothness hypotheses for g , as outlined in the subsequent proposition, provide sufficient conditions for our analysis.

Proposition 4.1. *Let $g : [0, 1] \rightarrow [0, 1]$ be a C^3 -diffeomorphism with $g(0) = g(1)$. Let $N \in \mathbb{N}$, $h = 1/(N+1)$. To simplify notations, we extend g as C^3 function in neighborhood of $[0, 1]$ and for h small enough, we set*

$$h_{-1/2} = 0 - g(-h), \quad h_{N+3/2} = g((N+2)h) - 1, \quad h_j = \frac{h_{j-1/2} + h_{j+1/2}}{2}, \quad j = \{0, N+1\}. \quad (4.1)$$

Choosing θ_0 and for $j \in \{0, \dots, N+1\}$

$$\theta_j = \theta_0 + x_j, \quad x_j = g(jh). \quad (4.2)$$

we have, for $j \in \{0, \dots, N\}$

$$\theta_{j+1} - \theta_j = x_{j+1} - x_j = h_{j+1/2}, \quad (4.3)$$

$$h_{j+1/2}(\theta_{j+1}h_{j+1} - \theta_jh_j) = h^3 g'_{j+1/2} \left((\theta_0 + g_{j+1/2})g''_{j+1/2} + (g'_{j+1/2})^2 \right) + \mathcal{O}(h^4), \quad (4.4)$$

where we have used the notations

$$g_{j+1/2} = g((j+1/2)h), \quad g'_{j+1/2} = g'((j+1/2)h), \quad g''_{j+1/2} = g''((j+1/2)h). \quad (4.5)$$

we also have, for $j \in \{1, \dots, N\}$,

$$\frac{\theta_j h_j + \theta_{j+1} h_{j+1}}{h_{j+1/2}} - \frac{\theta_{j-1} h_{j-1} + \theta_j h_j}{h_{j+1/2}} = 2h_j + \mathcal{O}(h^2). \quad (4.6)$$

The functions g_θ , $\theta > 0$ given in (3.4) which demonstrably satisfy the conditions outlined in Proposition 4.1, are precisely those employed in establishing the estimate (3.1).

4.1. Technical lemmas

Poincaré inequality tailored for non-uniform meshes. We then present three intermediate Lemmas, which serve as essential steps toward proving the estimate (3.1). Proposition 4.1 is utilized solely in the proof of Lemma 4.7.

To establish uniform observability estimates, it is crucial to control the low-frequency components of the discrete solution. The discrete Poincaré inequality below provides this control by linking the discrete L_2 -norm of the solution to the discrete gradient, which can be further related to boundary terms involving high-frequency modes. This inequality ensures that low-frequency modes cannot concentrate away from the boundary, thus playing a fundamental role in the overall observability analysis.

Proposition 4.2. *Let $N \in \mathbb{N}^*$ and $h = 1/(N+1)$. For any sufficiently smooth diffeomorphism on $[0, 1]$, satisfying $g(0) = 0$ and $g(1) = 0$, there exists $\alpha_0 > 0$ independent of h such that the following holds:*

$$\sum_{j=1}^N h_j |u_j|^2 \leq \alpha_0 \sum_{j=0}^N h_{j+1/2} \left| \frac{u_{j+1} - u_j}{h_{j+1/2}} \right|^2, \quad (4.7)$$

for any $u \in \mathbb{R}^N$, under the auxiliary condition $u_0 = u_{N+1} = 0$, where $h_{j+1/2}$ and h_j are quantities defined as in (2.2) and (2.3).

Proof. Let $u \in \mathbb{R}^N$, such that $u_0 = u_{N+1} = 0$, we have

$$h_j |u_j|^2 = h_j \left(\sum_{k=1}^j (u_k - u_{k-1}) \right)^2. \quad (4.8)$$

Thus, by using Schwarz's inequality, we obtain

$$\begin{aligned} \sum_{j=1}^N h_j |u_j|^2 &= \sum_{j=1}^N h_j \left(\sum_{k=1}^j (u_k - u_{k-1}) \right)^2 \\ &\leq \sum_{j=1}^N j h_j \sum_{k=1}^j (u_k - u_{k-1})^2. \end{aligned} \quad (4.9)$$

The following Taylor expansion is employed

$$h_{j+1/2} = h g'_j + g''_j \frac{h^2}{2} + \mathcal{O}(h^3), \quad j = 0, \dots, N+1, \quad (4.10)$$

$$h_{j-1/2} = h g'_j - g''_j \frac{h^2}{2} + \mathcal{O}(h^3), \quad j = 1, \dots, N, \quad (4.11)$$

$$h_j = \frac{h_{j+1/2} + h_{j-1/2}}{2} = 2hg'_j + \mathcal{O}(h^3), \quad j = 1, \dots, N+1. \quad (4.12)$$

It follows from (4.9) and (4.12) that there exists a constant $C_1 > 0$ independent of h , such that:

$$\begin{aligned} \sum_{j=1}^N h_j |u_j|^2 &\leq C_1 h \sum_{j=1}^N j \sum_{k=1}^j (u_k - u_{k-1})^2 \\ &\leq C_1 \sum_{j=1}^N \sum_{k=1}^j (u_k - u_{k-1})^2 \\ &= C_1 \sum_{j=1}^N (N-j+1)(u_j - u_{j-1})^2 \\ &= C_1 \sum_{j=1}^N (N-j+1) |h_{j-1/2}|^2 \left| \frac{u_j - u_{j-1}}{h_{j-1/2}} \right|^2, \end{aligned} \quad (4.13)$$

There exists $C_2 > 0$ independent of h , such that:

$$(N-j+1)h_{j-1/2} \leq C_2(N-j+1)h \leq C_2(N+1)h = C_2. \quad (4.14)$$

The proof follows directly from the above estimates. $\square$

Remark 4.3. In the above Poincaré estimate, the C^1 smoothness of the diffeomorphism g is sufficient. Moreover, The constant involved in the discrete Poincaré inequality (4.7) is evidently not optimal. The optimal constants can be derived using a spectral approach. However, in our setting, the spectrum is not explicitly available, making this approach difficult to apply.

The following statements deal with the conservation of the energy.

Proposition 4.4. Let $\alpha \neq 0$ and considering g as any diffeomorphism on $[0, 1]$ with $g(0) = 0$ and $g(1) = 1$, then for any solution (φ_h, ψ_h) of system (2.4), we have

$$\mathcal{E}_h(t) = \mathcal{E}_h(0), \quad \forall t > 0. \quad (4.15)$$

Proof. Multiplying the first equation of (2.4) by φ'_j we get

$$\sum_{j=1}^N h_j \varphi''_j \varphi'_j + \sum_{j=0}^N \frac{\varphi'_{j+1} - \varphi'_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} + \alpha \sum_{j=1}^N h_j \psi'_j \varphi'_j = 0.$$

Equivalently

$$\frac{1}{2} \frac{d}{dt} \sum_{j=1}^N h_j |\varphi'_j|^2 + \frac{1}{2} \frac{d}{dt} \sum_{j=0}^N h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 + \alpha \sum_{j=1}^N h_j \psi'_j \varphi'_j = 0. \quad (4.16)$$

Similarly, multiplying the second equation of (2.11) by ψ'_h , we get

$$\frac{1}{2} \frac{d}{dt} \sum_{j=1}^N h_j |\psi'_j|^2 + \frac{1}{2} \frac{d}{dt} \sum_{j=0}^N h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 - \alpha \sum_{j=1}^N h_j \psi'_j \varphi'_j = 0. \quad (4.17)$$

Combining Eqs. (4.16) and (4.17), we obtain

$$\frac{d}{dt} \mathcal{E}_h(t) = 0.$$

This ends the proof. $\square$

Lemma 4.5 delivers a crucial energy-exchange estimate between the two components of the semi-discrete system (2.4). It bounds the energy of φ_h by the energy of ψ_h plus the initial total energy, a fact that will be instrumental in our subsequent analysis.

Lemma 4.5. Let $\alpha \neq 0$ and $T > 0$, and considering g as any diffeomorphism on $[0, 1]$ with $g(0) = 0$ and $g(1) = 1$, then the following estimate holds for any initial state $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ of the system (2.4):

$$\alpha \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt \leq \alpha \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + 2\mathcal{E}_h(0), \quad (4.18)$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.4).

Proof. Multiplying the second equation of system (2.4) by φ'_j , we obtain

$$\sum_{j=1}^N \int_0^T h_j \psi''_j \varphi'_j dt - \sum_{j=1}^N \int_0^T \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} - \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) \varphi'_j dt - \alpha \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt = 0,$$

Integrating by parts, we get

$$\sum_{j=1}^N h_j \psi'_j \varphi'_j \Big|_0^T + \sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} \Big|_0^T - \sum_{j=0}^N \int_0^T h_j \psi'_j \varphi''_j dt \quad (4.19)$$

$$-\sum_{j=0}^N \int_0^T \frac{\psi'_{j+1} - \psi'_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} dt - \alpha \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt = 0.$$

Multiplying the first equation of (2.4) by ψ'_j , one has

$$\sum_{j=1}^N \int_0^T h_j \varphi''_j \psi'_j dt + \sum_{j=0}^N \int_0^T \frac{\psi'_{j+1} - \psi'_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} dt + \alpha \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt = 0. \quad (4.20)$$

By adding (4.19) to (4.20), we get

$$\alpha \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt = \alpha \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \sum_{j=1}^N h_j \psi'_j \varphi'_j \Big|_0^T + \sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} \Big|_0^T. \quad (4.21)$$

For all $t \in (0, T)$, we have

$$\sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} \leq \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 + \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2.$$

Likewise, we find that

$$\sum_{j=1}^N h_j \psi'_j \varphi'_j \leq \frac{1}{2} \sum_{j=1}^N h_j |\varphi'_j|^2 + \frac{1}{2} \sum_{j=1}^N h_j |\psi'_j|^2.$$

Thus

$$\sum_{j=1}^N h_j \psi'_j \varphi'_j \Big|_0^T + \sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} \Big|_0^T \leq 2\mathcal{E}_h(0). \quad (4.22)$$

From Eq. (4.21) and inequality (4.22), the proof is complete. $\square$

Lemma 4.6. Let $\alpha > 0$ and $T > 0$, and considering g as any diffeomorphism on $[0, 1]$ with $g(0) = 0$ and $g(1) = 1$, then the following estimate holds for any initial state $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ of the system (2.4):

$$\begin{aligned} & -\sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \left(1 - \frac{\alpha\alpha_0}{2}\right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & - \frac{\alpha h}{2} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt \leq 2 \max(1, \alpha_0) \mathcal{E}_h(0), \end{aligned}$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.4) and α_0 is the discrete Poincaré's constant in (4.7).

Proof. Multiplying the first equation of system (2.4) by φ_j , we obtain

$$\sum_{j=1}^N \int_0^T h_j \varphi''_j \varphi_j dt + \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt + \alpha \sum_{j=1}^N \int_0^T h_j \psi'_j \varphi_j dt = 0.$$

By integrating by parts, we get

$$-\sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt = -\sum_{j=1}^N h_j \varphi'_j \varphi_j \Big|_0^T - \alpha \sum_{j=1}^N \int_0^T h_j \psi'_j \varphi_j dt. \quad (4.23)$$

Using the discrete Poincaré's inequality (4.7), we obtain for any $\alpha > 0$

$$\begin{aligned} \alpha \sum_{j=1}^N \int_0^T h_j \psi'_j \varphi_j dt & \leq \frac{\alpha}{2} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{\alpha}{2} \sum_{j=1}^N \int_0^T h_j |\varphi_j|^2 dt \\ & \leq \frac{\alpha}{2} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{\alpha\alpha_0}{2} \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt. \end{aligned} \quad (4.24)$$

Similarly, for all $t \in [0, T]$, we have

$$\sum_{j=1}^N h_j \varphi'_j \varphi_j \leq \frac{1}{2} \sum_{j=1}^N h_j |\varphi'_j|^2 dt + \frac{\alpha_0}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \leq \max(1, \alpha_0) \mathcal{E}_h(0).$$

Therefore

$$\sum_{j=1}^N h_j \varphi'_j \varphi_j \Big|_0^T \leq 2 \max(1, \alpha_0) \mathcal{E}_h(0). \quad (4.25)$$

The proof is completed by the combination of (4.23), (4.24), and (4.25). $\square$

Lemma 4.7 provides a crucial multiplier estimate showing how the specific choice of the parameter θ in the diffeomorphism g_θ induces a spectral gap. This gap is key to controlling high-frequency components of the solution through the boundary trace, thereby playing an essential role in the overall observability analysis.

Lemma 4.7. Let $\alpha > 0$, $T > 0$ and consider g_θ defined in (3.4), with $\theta > 0$ as a specific choice of the diffeomorphism g . There exist constants $C_1(\theta) > 0$ and $C_2 > 0$ (independent of h) such that, for any initial condition $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ associated with the system (2.4), the following estimate holds:

$$\begin{aligned} & \left[1 - \frac{\alpha(\theta + 1)}{2}\right] \sum_{j=1}^N \int_0^T h_j |\psi_j'|^2 dt + [1 - 2\alpha(\theta + 1)] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ & \leq (\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + [C_1(\theta) + C_2 h] \mathcal{E}_h(0), \end{aligned} \quad (4.26)$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.4). The constant $C_1(\theta)$ is explicitly given by:

$$C_1(\theta) = 9(\theta + 1). \quad (4.27)$$

Proof. Consider $m_j(t)$, the multiplier specified as follows:

$$m_j(t) = \theta_j \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} + \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right), \quad \text{with } \theta_j = \theta_0 + x_j. \quad (4.28)$$

where, θ_0 is an arbitrary parameter to be specified later. By multiplying the second equation of the system (2.4) by m_j and then applying integration by parts, we derive the following expression

$$\begin{aligned} & \sum_{j=1}^N h_j \psi_j' m_j \Big|_0^T - \underbrace{\sum_{j=1}^N \int_0^T h_j \psi_j' m_j' dt}_A \\ & - \underbrace{\sum_{j=1}^N \int_0^T \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} - \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) m_j dt}_{B} - \alpha \sum_{j=1}^N \int_0^T h_j \varphi_j' m_j dt = 0. \end{aligned} \quad (4.29)$$

The estimation of the term A relies on the Taylor expansions (4.4) and (4.6), presented in Proposition 4.1, and is determined as follows:

$$\begin{aligned} A &= \sum_{j=1}^N \int_0^T h_j \theta_j \left(\frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} + \frac{\psi_j' - \psi_{j-1}'}{h_{j-1/2}} \right) \psi_j' dt \\ &= \sum_{j=0}^N \int_0^T (h_j \theta_j \psi_j' + h_{j+1} \theta_{j+1} \psi_{j+1}') \left(\frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right) dt \\ &= \frac{1}{2} \sum_{j=0}^N \int_0^T (h_j \theta_j + h_{j+1} \theta_{j+1}) \left(\frac{|\psi_{j+1}'|^2 - |\psi_j'|^2}{h_{j+1/2}} \right) dt \\ &\quad + \frac{1}{2} \sum_{j=0}^N \int_0^T h_{j+1/2} (h_{j+1} \theta_{j+1} - h_j \theta_j) \left| \frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right|^2 dt \\ &= -\frac{1}{2} \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1} \theta_{j+1} + h_j \theta_j}{h_{j+2}} - \frac{h_j \theta_j + h_{j-1} \theta_{j-1}}{h_{j-1/2}} \right) |\psi_j'|^2 dt \\ &\quad + \frac{1}{2} \sum_{j=0}^N \int_0^T h_{j+1/2} (h_{j+1} \theta_{j+1} - h_j \theta_j) \left| \frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right|^2 dt \\ &= -\sum_{j=1}^N \int_0^T h_j |\psi_j'|^2 dt + \mathcal{O}(h^2) \sum_{j=1}^N \int_0^T |\psi_j'|^2 dt + \mathcal{O}(h^4) \sum_{j=0}^N \int_0^T \left| \frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right|^2 dt \\ &\quad + \frac{1}{2} \sum_{j=0}^N \int_0^T \left[h^3 g_{j+1/2}' \left((\theta_0 + g_{j+1/2}) g_{j+1/2}'' + (g_{j+1/2}')^2 \right) \right] \left| \frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right|^2 dt. \end{aligned} \quad (4.30)$$

The estimation of B is derived from Eq. (4.3) and is expressed as:

$$\begin{aligned} B &= \sum_{j=1}^N \int_0^T \theta_j \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} - \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} + \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) dt \\ &= \sum_{j=1}^N \int_0^T \theta_j \left(\left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 - \left| \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right|^2 \right) dt \\ &= -\sum_{j=0}^N \int_0^T (\theta_{j+1} - \theta_j) \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt + \theta_{N+1} \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt - \theta_0 \int_0^T \left| \frac{\psi_1}{h_{1/2}} \right|^2 dt \\ &= -\sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt + \theta_{N+1} \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt - \theta_0 \int_0^T \left| \frac{\psi_1}{h_{1/2}} \right|^2 dt. \end{aligned} \quad (4.31)$$

In view of (4.29), (4.30) and (4.31) we get

$$\begin{aligned} & \sum_{j=1}^N h_j \psi_j' m_j \Big|_0^T + \sum_{j=1}^N \int_0^T h_j |\psi_j'|^2 dt + \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt + \mathcal{O}(h^2) \sum_{j=1}^N \int_0^T |\psi_j'|^2 dt \\ & - \frac{1}{2} \sum_{j=0}^N \int_0^T \left[h^3 g_{j+1/2}' \left((\theta_0 + g_{j+1/2}) g_{j+1/2}'' + (g_{j+1/2}')^2 \right) \right] \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ & + \mathcal{O}(h^4) \sum_{j=0}^N \int_0^T \left| \frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right|^2 dt - \alpha \sum_{j=1}^N \int_0^T h_j \varphi_j' m_j dt \\ & = \theta_{N+1} \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt - \theta_0 \int_0^T \left| \frac{\psi_1}{h_{1/2}} \right|^2 dt. \end{aligned} \quad (4.32)$$

On the other hand, it can be observed that there exists a constant $\hat{C}_1 > 0$, independent of h , such that the following estimate is satisfied:

$$\mathcal{O}(h^2) \sum_{j=1}^N \int_0^T |\psi_j'|^2 dt + \mathcal{O}(h^4) \sum_{j=0}^N \int_0^T \left| \frac{\psi_{j+1}' - \psi_j'}{h_{j+1/2}} \right|^2 dt \leq \hat{C}_1 h \mathcal{E}_h(0). \quad (4.33)$$

Furthermore, using the fact that $\theta_j \leq 1 + \theta_0$, we can deduce the following for any $\alpha > 0$:

$$\begin{aligned} \alpha \sum_{j=1}^N \int_0^T h_j \varphi_j' m_j dt &= \alpha \sum_{j=1}^N \int_0^T h_j \theta_j \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} + \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) \varphi_j' dt \\ &\leq \frac{\alpha(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T h_j |\varphi_j'|^2 dt + \frac{\alpha(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T h_j \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} + \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right|^2 dt \\ &\leq \frac{\alpha(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T h_j |\varphi_j'|^2 dt + \alpha(1 + \theta_0) \sum_{j=0}^N \int_0^T (h_{j+1} + h_j) \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt. \end{aligned} \quad (4.34)$$

In light of Eq. (4.1), we have

$$\frac{h_{j+1} + h_j}{h_{j+1/2}} = 2 + \mathcal{O}(h^2). \quad (4.35)$$

By using Eq. (4.35) and the estimate from Lemma 4.5, we obtain

$$\begin{aligned} \alpha \sum_{j=1}^N \int_0^T h_j \varphi_j' m_j dt &\leq \frac{\alpha(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T h_j |\varphi_j'|^2 dt \\ &\quad + \alpha(1 + \theta_0) \sum_{j=0}^N \int_0^T [2h_{j+1/2} + \mathcal{O}(h^3)] \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ &\leq \frac{\alpha(\theta_0 + 1)}{2} \sum_{j=1}^N \int_0^T h_j |\psi_j'|^2 dt + 2\alpha(\theta_0 + 1) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ &\quad + [1 + \theta_0 + \mathcal{O}(h^2)] \mathcal{E}_h(0). \end{aligned} \quad (4.36)$$

Following the same techniques used to obtain estimation (4.34), we get

$$\begin{aligned} \sum_{j=1}^N h_j \psi_j' m_j &= \sum_{j=1}^N h_j \theta_j \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} + \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) \psi_j' dt \\ &\leq \frac{(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T h_j |\psi_j'|^2 dt + (1 + \theta_0) \sum_{j=0}^N \int_0^T (h_{j+1} + h_j) \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ &\leq [4(1 + \theta_0) + \mathcal{O}(h^2)] \mathcal{E}_h(0). \end{aligned} \quad (4.37)$$

A straightforward calculation verifies that for $x \in [0, 1]$, the following holds:

$$\frac{g_\theta'(x)^2 + g_\theta(x) g_\theta''(x)}{-g_\theta''(x)} = \theta.$$

By setting $\theta = \theta_0$, in the multiplier defined in (4.28), it follows that:

$$h^3 g_{j+1/2}' \left((\theta_0 + g_{j+1/2}) g_{j+1/2}'' + (g_{j+1/2}')^2 \right) = 0, \quad j = 0, \dots, N. \quad (4.38)$$

Applying the results from (4.33), (4.36), (4.37), and (4.38) to (4.32) yields the desired outcome:

$$\begin{aligned} & \left[1 - \frac{\alpha(\theta + 1)}{2} \right] \sum_{j=1}^N \int_0^T h_j |\psi_j'|^2 dt + [1 - 2\alpha(\theta + 1)] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ & \leq (\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + [9(1 + \theta) + \mathcal{O}(h)] \mathcal{E}_h(0). \quad \square \end{aligned} \quad (4.39)$$

4.2. Proof of observability inequality (3.1)

We are now in a position to prove the estimate in [Theorem 3.1](#). As previously mentioned, the proof of the theorem relies on the diffeomorphism (3.4). [Lemma 4.6](#) can be equivalently written as:

$$\begin{aligned} & \frac{1}{2} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \left(1 - \frac{\alpha\alpha_0}{2}\right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & - \frac{3}{2} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt - \frac{\alpha h}{2} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt \leq 2 \max(1, \alpha_0) \mathcal{E}_h(0), \end{aligned} \quad (4.40)$$

Combining estimate (4.40) with the estimate from [Lemma 4.5](#), we get:

$$\begin{aligned} & \frac{1}{2} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \left(1 - \frac{\alpha\alpha_0}{2}\right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt - \frac{1}{2} (3 + \alpha) \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt \\ & \leq \left[2 \max(1, \alpha_0) + \frac{3}{\alpha} \right] \mathcal{E}_h(0), \end{aligned} \quad (4.41)$$

Now, by adding estimate (4.41) to twice the estimate from [Lemma 4.7](#), we obtain:

$$\begin{aligned} & \frac{1}{2} [1 - \alpha(2\theta + 3)] \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + 2[1 - 2\alpha(\theta + 1)] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ & + \frac{1}{2} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \left(1 - \frac{\alpha\alpha_0}{2}\right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & \leq 2(\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + [2C_1(\theta) + 2C_2h + 2 \max(1, \alpha_0) + \frac{3}{\alpha}] \mathcal{E}_h(0), \end{aligned} \quad (4.42)$$

By applying the energy conservation property (5.7), the above inequality leads to:

$$\begin{aligned} & \frac{1}{2} [1 - 3\alpha] \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + 2[1 - 2\alpha] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ & + \frac{1}{2} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \left(1 - \frac{\alpha\alpha_0}{2}\right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & \leq 2(\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + [2C_1(\theta) + 8T\alpha\theta + 2C_2h + 2 \max(1, \alpha_0) + \frac{3}{\alpha}] \mathcal{E}_h(0). \end{aligned} \quad (4.43)$$

Let $M(\alpha)$ be defined for α in the interval $[0, \hat{\alpha}]$, as follows:

$$M(\alpha) = \min \left[1, (1 - 3\alpha), 4(1 - 2\alpha), (2 - \alpha\alpha_0) \right], \quad 0 < \alpha \leq \hat{\alpha} := \min \left[\frac{1}{3}, \frac{2}{\alpha_0}, \frac{1}{2} \right], \quad (4.44)$$

and let $\hat{C}$ be the constant multiplier in the energy in (4.43). Then, the following inequality holds:

$$TM(\alpha) \mathcal{E}_h(0) - \hat{C}(\alpha, \theta, h) \mathcal{E}_h(0) \leq 2(\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt. \quad (4.45)$$

Now, a special and optimal treatment is required to recover the observability inequality (4.43) for the minimal time threshold possible. To this end, let $T_0(r)$ be the affine function defined for any $r > 0$ as follows:

$$T_0(r) = \left[r + 2 \max(1, \alpha_0) + \frac{3}{\alpha} \right] / M(\alpha). \quad (4.46)$$

Let ε in $(0, 1)$, such that the following two inequalities hold:

$$8\alpha\theta \leq \varepsilon M(\alpha), \quad 2C_1(\theta) + 2C_2h + 2 \max(1, \alpha_0) + \frac{3}{\alpha} \leq (1 - \varepsilon) M(\alpha) T_0(r). \quad (4.47)$$

In light of (4.45) and (4.47), for a given time T that exceeds the threshold $T_0(r)$, the following inequality holds:

$$\mathcal{E}_h(0) \leq C_r(\alpha, T, \theta) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt,$$

where

$$C_r(\alpha, T, \theta) := \frac{2(\theta + 1)}{(1 - \varepsilon) M(\alpha) (T - T_0(r))}.$$

The proof of [Theorem 3.1](#) is concluded if the second equation in (4.47) is satisfied. This condition is met for sufficiently large $r > 0$ and sufficiently small h . Moreover, the constant in the estimate (3.1) and the threshold time T_0 in [Theorem 3.1](#) are determined by choosing the pair (r, θ) , where r is minimized while ensuring θ remains within the largest possible interval.

Remark 4.8.

- The observability inequality (1.2) was demonstrated in Ref. 3 for time intervals T exceeding a threshold time $T_0 := T_0(r)$, with $r = 10$. However, for sufficiently small positive values of α , where multiplier techniques are employed, this threshold time T_0 has not been proven to be minimal in the strict sense of control theory.

- The discrete setting follows a careful adaptation of techniques from the continuous case, but the discrete observability inequality (3.1) is established only for times exceeding the threshold $T_0(r)$, which is strictly greater than T_0 . This restriction arises from the estimation in Lemma 4.7, where the involved constants deviate from their continuous counterparts through values of the parameter θ . The deviation primarily stems from the estimation in (4.36), where the employed technique appears to be the only viable option.
- The observability inequality (3.1) holds for $T > T_0(r)$ when r satisfies the second equality in (4.47), specifically for values of r exceeding $r_{\varepsilon,h}^*(\theta)$, which is given by

$$r_{\varepsilon,h}^*(\theta) := \frac{\varepsilon[2\max(1, \alpha_0) + (3/\alpha)] + 2C_2h + 2C_1(\theta)}{1 - \varepsilon}, \quad \theta \in (0, \varepsilon M(\alpha)/8\alpha). \quad (4.48)$$

- In the single wave case,³⁰ for any time $T > 2$, the parameter θ can be freely chosen within the interval $(0, (T-2)/2)$. However, due to the coupled nature of the system, this flexibility does not persist, as θ becomes constrained to the interval $(0, \varepsilon M(\alpha)/8\alpha)$. Furthermore, this interval for θ becomes increasingly restricted as r^* approaches its minimum possible values. Such limitation appears natural given that the analysis focuses on small values of α .

5. Proof of Theorem 3.2

This section focuses on recovering the observability inequality (3.2) uniformly with respect to h . The smoothness properties of g are assumed to be the same as in the finite differences context. The results stated in the following proposition, which are proven in Ref. 30, are technical and specifically address these issues.

Proposition 5.1. *Let $g : [0, 1] \rightarrow [0, 1]$ be a C^3 -diffeomorphism with $g(0) = 0$, $g(1) = 1$. Let $N \in \mathbb{N}$, $h = 1/(N+1)$. To simplify notations, we extend g as C^3 function in a neighborhood of $[0, 1]$ and for h small enough, we set $h_{-1/2}$, h_0 , $h_{N+3/2}$ and h_{N+1} as in (4.1). We choose θ_0 and for $j \in \{0, \dots, N+1\}$, we set θ_j as in (4.2). Using the same notations as in Propositions 4.1 and following notations*

$$g_j = g(jh), \quad g'_j = g'(jh), \quad g''_j = g''(jh), \quad (5.1)$$

we get the straightforward identities

$$\theta_j > 0, \quad \theta_{N+1} = 1 + \theta_0, \quad \theta_{j+1} - \theta_j = h_{j+1/2}, \quad j \in \{0, \dots, N+1\}, \quad (5.2)$$

and following Taylor expansions:

$$\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} = -\frac{2g''_j}{g'_j} + \mathcal{O}(h^2), \quad (5.3)$$

$$\theta_{j+1} \left(\frac{h_{j+1/2}}{h_{j+3/2}} + \frac{2h_{j+3/2}}{h_{j+1/2}} - 1 \right) + \theta_j \left(1 - \frac{2h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) = \frac{2h}{g'_{j+1/2}} \left((\theta_0 + g_{j+1/2})g''_{j+1/2} + (g'_{j+1/2})^2 \right) + \mathcal{O}(h^2), \quad (5.4)$$

$$\theta_{j+1} \left(\frac{3h_{j+3/2}}{h_{j+1/2}} - 1 \right) + 3\theta_j \left(\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) - \theta_{j-1} \left(\frac{3h_{j-3/2}}{h_{j-1/2}} - 1 \right) = 4h_j + \mathcal{O}(h^2). \quad (5.5)$$

The function g_θ represents the smoothness condition described in the above proposition and is crucially used in the proof of the uniform estimate (3.2). Given the smoothness and strict concavity of (3.4), the identity (5.3) simplifies to Ref. 30:

$$\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \geq 0, \quad j \in \{0, \dots, N+1\}. \quad (5.6)$$

5.1. Technical lemmas

Proposition 5.2. *Let $\alpha \neq 0$ and considering g as any diffeomorphism on $[0, 1]$ with $g(0) = 0$ and $g(1) = 1$, then for any solution (φ_h, ψ_h) of system (2.11), we have*

$$\mathcal{E}_h(t) = \mathcal{E}_h(0), \quad \forall t > 0. \quad (5.7)$$

Proof. Multiplying the first equation of (2.11) by φ'_j we get

$$\begin{aligned} & \frac{1}{6} \sum_{j=0}^N h_{j+1/2} (\varphi''_j + \varphi''_{j+1}) (\varphi'_j + \varphi'_{j+1}) + \frac{1}{3} \sum_{j=1}^N h_j \varphi''_j \varphi'_j \\ & + \sum_{j=0}^N \frac{\varphi'_{j+1} - \varphi'_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} + \alpha \sum_{j=1}^N h_j \psi'_j \varphi'_j = 0 \end{aligned}$$

Equivalently

$$\begin{aligned} & \frac{1}{12} \frac{d}{dt} \sum_{j=0}^N h_{j+1/2} |\varphi'_j + \varphi'_{j+1}|^2 + \frac{1}{6} \frac{d}{dt} \sum_{j=1}^N h_j |\varphi'_j|^2 \\ & + \frac{1}{2} \frac{d}{dt} \sum_{j=0}^N h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 + \alpha \sum_{j=1}^N h_j \psi'_j \varphi'_j = 0. \end{aligned} \quad (5.8)$$

Similarly, multiplying the second equation of (2.11) by ψ'_j , we get

$$\begin{aligned} & \frac{1}{12} \frac{d}{dt} \sum_{j=0}^N h_{j+1/2} |\psi'_j + \psi'_{j+1}|^2 + \frac{1}{6} \frac{d}{dt} \sum_{j=1}^N h_j |\psi'_j|^2 \\ & + \frac{1}{2} \frac{d}{dt} \sum_{j=0}^N h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 - \alpha \sum_{j=1}^N h_j \psi'_j \varphi'_j = 0. \end{aligned} \quad (5.9)$$

Combining Eqs. (5.8) and (5.9), we obtain

$$\frac{d}{dt} \mathcal{E}_h(t) = 0.$$

This ends the proof. $\square$

Lemma 5.3. Let $\alpha \neq 0$ and $T > 0$, and considering g as any diffeomorphism on $[0, 1]$ with $g(0) = 0$ and $g(1) = 1$, then the following estimate holds for any initial state $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ of the system (2.11):

$$\begin{aligned} & \frac{\alpha}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \frac{\alpha}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt \\ & \leq \frac{\alpha}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{\alpha}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt + 2\mathcal{E}_h(0), \end{aligned} \quad (5.10)$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.11).

Proof. By multiplying the second equation of system (2.11) by φ'_j and integrating by parts, we use the self-adjoint property of the mass matrix to obtain

$$\begin{aligned} & \sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi'_j \Big|_0^T + \sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} \Big|_0^T \\ & - \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \varphi''_{j+1} + \frac{2h_j}{3} \varphi''_j + \frac{h_{j-1/2}}{6} \varphi''_{j-1} \right) \psi'_j dt - \sum_{j=0}^N \int_0^T \frac{\psi'_{j+1} - \psi'_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} dt \\ & - \frac{\alpha}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt - \frac{\alpha}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt = 0 \end{aligned} \quad (5.11)$$

By multiplying the first equation of (2.11) by ψ'_j , we obtain

$$\begin{aligned} & \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \varphi''_{j+1} + \frac{2h_j}{3} \varphi''_j + \frac{h_{j-1/2}}{6} \varphi''_{j-1} \right) \psi'_j dt + \sum_{j=0}^N \int_0^T \frac{\psi'_{j+1} - \psi'_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} dt \\ & + \frac{\alpha}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{\alpha}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt = 0 \end{aligned} \quad (5.12)$$

Adding (5.11) to (5.12), we get

$$\begin{aligned} & \frac{\alpha}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \frac{\alpha}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt \\ & = \frac{\alpha}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{\alpha}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \\ & + \sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi'_j \Big|_0^T + \sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{\sqrt{h_{j+1/2}}} \Big|_0^T \end{aligned} \quad (5.13)$$

For all $t \in (0, T)$, we have

$$\sum_{j=1}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \leq \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 + \frac{1}{2} \sum_{j=0}^N h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2$$

Additionally, for all $t \in (0, T)$, we have

$$\begin{aligned} & \sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{3} \psi'_{j-1} \right) \varphi'_j \\ & = \sum_{j=1}^N \frac{h_{j+1/2}}{6} (\psi'_{j+1} + \psi'_j) \varphi'_j + \frac{h_{j-1/2}}{6} (\psi'_j + \psi'_{j-1}) \varphi'_j \\ & = \frac{1}{6} \sum_{j=0}^N h_{j+1/2} (\psi'_{j+1} + \psi'_j) (\varphi'_{j+1} + \varphi'_j) \end{aligned}$$

$$\leq \frac{1}{12} \sum_{j=0}^N h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 + \frac{1}{12} \sum_{j=0}^N h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2.$$

It follows that

$$\sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi'_j \Big|_0^T + \sum_{j=0}^N \frac{\psi_{j+1} - \psi_j}{\sqrt{h_{j+1/2}}} \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \Big|_0^T \leq 2\mathcal{E}_h(0). \quad (5.14)$$

By considering the expressions referenced in (5.12) and (5.14), the proof is concluded. $\square$

Lemma 5.4. Let $\alpha > 0$ and $T > 0$, and considering g as any diffeomorphism on $[0, 1]$ with $g(0) = 0$ and $g(1) = 1$, then, there exists a constant $\hat{C}_1 > 0$ (independent of h) such that, for any initial state $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ of the system (2.11):

$$\begin{aligned} & - \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt \right) \\ & + \left(1 - \frac{\alpha \alpha_0}{2} \right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & - \frac{\alpha}{2} \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \right) \\ & \leq [2 \max(1, \alpha_0) + \hat{C}_1 h^2] \mathcal{E}_h(0), \end{aligned}$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.11) and α_0 is the discrete Poincaré's constant in (4.7).

Proof. By multiplying the first equation of system (2.11) by φ_j , we obtain

$$\begin{aligned} & \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \varphi''_{j+1} + \frac{2h_j}{3} \varphi''_j + \frac{h_{j-1/2}}{6} \varphi''_{j-1} \right) \varphi_j dt + \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & + \alpha \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi_j dt = 0. \end{aligned} \quad (5.15)$$

Integration by parts in (5.15) yields:

$$\begin{aligned} & -\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt - \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt + \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ & = - \sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \varphi'_{j+1} + \frac{2h_j}{3} \varphi'_j + \frac{h_{j-1/2}}{6} \varphi'_{j-1} \right) \varphi_j \Big|_0^T \\ & \quad - \alpha \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi_j dt. \end{aligned} \quad (5.16)$$

In view of Poincaré's inequality (4.7), we obtain:

$$\begin{aligned} & \alpha \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi_j dt \\ & \leq \frac{\alpha}{2} \sum_{j=1}^N \int_0^T \frac{1}{h_j} \left| \frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right|^2 dt + \frac{\alpha}{2} \sum_{j=1}^N \int_0^T h_j |\varphi_j|^2 dt \\ & \leq \frac{\alpha}{2} \sum_{j=1}^N \int_0^T \frac{1}{h_j} \left| \frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right|^2 dt + \frac{\alpha \alpha_0}{2} \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt. \end{aligned} \quad (5.17)$$

Applying Young's inequality and performing a Taylor expansion, we deduce the existence of a constant $\hat{C}_{1,1}$, independent of h , such that:

$$\begin{aligned} & \sum_{j=1}^N \frac{1}{h_j} \left| \frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right|^2 \\ & = \sum_{j=1}^N \frac{1}{h_j} \left(\frac{h_{j+1/2}}{6} (\psi'_{j+1} + \psi'_j) + \frac{h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} (\psi'_j + \psi'_{j-1}) \right)^2 \\ & \leq \frac{3}{2} \sum_{j=1}^N \frac{1}{h_j} \left(\frac{h_{j+1/2}}{6} (\psi'_{j+1} + \psi'_j) + \frac{h_{j-1/2}}{6} (\psi'_j + \psi'_{j-1}) \right)^2 + 3 \sum_{j=1}^N \frac{h_j}{9} |\psi'_j|^2 \\ & \leq 3 \sum_{j=1}^N \frac{1}{h_j} \left(\frac{h_{j+1/2}}{6} (\psi'_{j+1} + \psi'_j) \right)^2 + 3 \sum_{j=1}^N \frac{1}{h_j} \left(\frac{h_{j-1/2}}{6} (\psi'_j + \psi'_{j-1}) \right)^2 + \frac{1}{3} \sum_{j=1}^N h_j |\psi'_j|^2 \\ & \leq \frac{1}{6} \sum_{j=0}^N h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 + \frac{1}{3} \sum_{j=1}^N h_j |\psi'_j|^2 + \hat{C}_{1,1} h^2 \mathcal{E}_h(0). \end{aligned} \quad (5.18)$$

Integrating the estimate (5.18), applying the energy conservation property (5.7), and subsequently using (5.17), yields:

$$\begin{aligned} & \alpha \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) \varphi_j dt \\ & \leq \frac{\alpha}{12} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt + \frac{\alpha}{6} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt \\ & + \frac{\alpha \alpha_0}{2} \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt + \alpha T \hat{C}_{1,1} h^2 \mathcal{E}_h(0). \end{aligned} \quad (5.19)$$

Applying the same technique used in the estimation of (5.19), along with the energy conservation property, we obtain:

$$\sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \varphi'_{j+1} + \frac{2h_j}{3} \varphi'_j + \frac{h_{j-1/2}}{6} \varphi'_{j-1} \right) \varphi_j \Big|_0^T \leq (2 \max(1, \alpha_0) + \hat{C}_{1,1} h^2) \mathcal{E}_h(0). \quad (5.20)$$

In light of (5.15), (5.19), and (5.20), this completes the proof. $\square$

Lemma 5.5. Let $\alpha > 0$, $T > 0$ and consider g_θ defined in (3.4), with $\theta > 0$ as a specific choice of the diffeomorphism g . There exist constants $C_1(\theta) > 0$ and $\hat{C}_2 > 0$ (independent of h) such that, for any initial condition $(\varphi_h^0, \varphi_h^1, \psi_h^0, \psi_h^1) \in (\mathbb{R}^N \times \mathbb{R}^N)^2$ associated with the system (2.11), the following estimate holds:

$$\begin{aligned} & \left[1 - \frac{\alpha(\theta+1)}{2} \right] \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \right) \\ & + [1 - 2\alpha(\theta+1)] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ & \leq (\theta+1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + \frac{\theta+1}{6} \int_0^T |\psi'_N|^2 dt + [C_1(\theta) + \hat{C}_2 h] \mathcal{E}_h(0), \end{aligned} \quad (5.21)$$

where (φ_h, ψ_h) is the corresponding unique solution of (2.11). The constant $C_1(\theta)$ is explicitly given by:

$$C_1(\theta) = 9(\theta+1). \quad (5.22)$$

Proof. Multiplying the second equation of (2.11) by the multiplier given in (4.28) and integrating by parts, we obtain

$$\begin{aligned} & \sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) m_j \Big|_0^T - \underbrace{\sum_{j=1}^N \int_0^T \left(\frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} - \frac{\psi_j - \psi_{j-1}}{h_{j-1/2}} \right) m_j dt}_B \\ & - \underbrace{\sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) m'_j dt - \alpha \sum_{j=1}^N \int_0^T h_j \varphi'_j m_j dt}_A = 0 \end{aligned} \quad (5.23)$$

In the context of a single wave,³⁰ the term A is well-established and can be represented as the sum of four terms, expressed as follows:

$$A = A_1 + A_2 + A_3 + A_4, \quad (5.24)$$

where,

$$A_1 = \frac{\theta_{N+1}}{6} \int_0^T |\psi'_N|^2 dt - \frac{\theta_0}{6} \int_0^T |\psi'_1|^2 dt - \frac{1}{6} \sum_{j=0}^N \int_0^T (\theta_{j+1} - \theta_j) |\psi'_{j+1} + \psi'_j|^2 dt, \quad (5.25)$$

$$A_2 = -\frac{1}{12} \sum_{j=1}^N \int_0^T \theta_j \left(\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) |\psi'_{j+1} - \psi'_{j-1}|^2 dt, \quad (5.26)$$

$$\begin{aligned} A_3 = & -\frac{\theta_0}{12} \left(\frac{h_{-1/2}}{h_{1/2}} - \frac{h_{1/2}}{h_{-1/2}} \right) \int_0^T |\psi'_1|^2 dt - \frac{\theta_{N+1}}{12} \left(\frac{h_{N+1/2}}{h_{N+3/2}} - \frac{h_{N+3/2}}{h_{N+1/2}} \right) \int_0^T |\psi'_N|^2 dt \\ & + \frac{1}{12} \sum_{j=0}^N \int_0^T \left[\theta_{j+1} \left(\frac{h_{j+1/2}}{h_{j+3/2}} - \frac{h_{j+3/2}}{h_{j+1/2}} \right) + \theta_j \left(\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) \right] |\psi'_{j+1} - \psi'_j|^2 dt, \end{aligned} \quad (5.27)$$

$$\begin{aligned} A_4 = & -\frac{1}{12} \sum_{j=1}^N \int_0^T \left[\theta_{j+1} \left(\frac{3h_{j+3/2}}{h_{j+1/2}} - 1 \right) + 3\theta_j \left(\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) - \theta_{j-1} \left(\frac{3h_{j-3/2}}{h_{j-1/2}} - 1 \right) \right] |\psi'_j|^2 dt \\ & + \frac{1}{12} \sum_{j=0}^N \int_0^T \left[\theta_{j+1} \left(\frac{3h_{j+3/2}}{h_{j+1/2}} - 1 \right) - \theta_j \left(\frac{3h_{j-1/2}}{h_{j+1/2}} - 1 \right) \right] |\psi'_{j+1} - \psi'_j|^2 dt. \end{aligned} \quad (5.28)$$

we have

$$-A_1 \geq -\frac{\theta_{N+1}}{6} \int_0^T |\psi'_N|^2 dt + \frac{1}{6} \sum_{j=1}^N \int_0^T (\theta_{j+1} - \theta_j) |\psi'_{j+1} + \psi'_j|^2 dt \quad (5.29)$$

The two properties (5.4) and (5.5) as outlined in Proposition 5.1, we derive the following:

$$\begin{aligned}
 -(\mathbf{A}_3 + \mathbf{A}_4) &= \frac{\theta_0}{12} \left(\frac{h_{-1/2}}{h_{1/2}} - \frac{h_{1/2}}{h_{-1/2}} \right) \int_0^T |\psi'_1|^2 dt + \frac{\theta_{N+1}}{12} \left(\frac{h_{N+1/2}}{h_{N+3/2}} - \frac{h_{N+3/2}}{h_{N+1/2}} \right) \int_0^T |\psi'_N|^2 dt \\
 &\quad - \frac{1}{12} \sum_{j=0}^N \int_0^T \left[\theta_{j+1} \left(\frac{h_{j+1/2}}{h_{j+3/2}} + \frac{2h_{j+3/2}}{h_{j+1/2}} - 1 \right) + \theta_j \left(1 - \frac{2h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) \right] |\psi'_{j+1} - \psi'_j|^2 dt \\
 &\quad + \frac{1}{12} \sum_{j=1}^N \int_0^T \left[\theta_{j+1} \left(\frac{3h_{j+3/2}}{h_{j+1/2}} - 1 \right) + 3\theta_j \left(\frac{h_{j-1/2}}{h_{j+1/2}} - \frac{h_{j+1/2}}{h_{j-1/2}} \right) - \theta_{j-1} \left(\frac{3h_{j-3/2}}{h_{j-1/2}} - 1 \right) \right] |\psi'_j|^2 dt \\
 &= \frac{\theta_0}{12} \left(\frac{h_{-1/2}}{h_{1/2}} - \frac{h_{1/2}}{h_{-1/2}} \right) \int_0^T |\psi'_1|^2 dt + \frac{\theta_{N+1}}{12} \left(\frac{h_{N+1/2}}{h_{N+3/2}} - \frac{h_{N+3/2}}{h_{N+1/2}} \right) \int_0^T |\psi'_N|^2 dt \\
 &\quad - \frac{1}{12} \sum_{j=1}^N \int_0^T \left[\frac{2h}{g'_{j+1/2}} \left((\theta_0 + g_{j+1/2}) g''_{j+1/2} + (g'_{j+1/2})^2 \right) + \mathcal{O}(h^2) \right] |\psi'_{j+1} - \psi'_j|^2 dt \\
 &\quad + \frac{1}{12} \sum_{j=1}^N \int_0^T (4h_j + \mathcal{O}(h^2)) |\psi'_j|^2 dt.
 \end{aligned} \tag{5.30}$$

Applying the positivity property (5.6) in $\mathbf{A}_2$ and in $\mathbf{A}_3 + \mathbf{A}_4$, along with the estimate (5.29), the term $\mathbf{A}$ is estimated as follows:

$$\begin{aligned}
 -\mathbf{A} &\geq -\frac{1}{12} \sum_{j=0}^N \int_0^T \left[\frac{2h}{g'_{j+1/2}} \left((\theta_0 + g_{j+1/2}) g''_{j+1/2} + (g'_{j+1/2})^2 \right) + \mathcal{O}(h^2) \right] |\psi'_{j+1} - \psi'_j|^2 dt \\
 &\quad + \frac{1}{6} \sum_{j=0}^N \int_0^T (\theta_{j+1} - \theta_j) |\psi'_{j+1} + \psi'_j|^2 dt + \frac{1}{12} \sum_{j=1}^N \int_0^T (4h_j + \mathcal{O}(h^2)) |\psi'_j|^2 dt \\
 &\quad - \frac{\theta_{N+1}}{6} \int_0^T |\psi'_N|^2 dt.
 \end{aligned} \tag{5.31}$$

The term $\mathbf{B}$ was previously computed in the context of the finite difference method, and we recall it here as

$$\mathbf{B} = - \sum_{j=0}^N \int_0^T (\theta_{j+1} - \theta_j) \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 + \theta_{N+1} \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt - \theta_0 \int_0^T \left| \frac{\psi_1}{h_{1/2}} \right|^2 dt. \tag{5.32}$$

In view of (5.23), (5.31), and (5.32), the following inequality holds

$$\begin{aligned}
 &\sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) m_j \Big|_0^T + \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\
 &\quad - \frac{1}{12} \sum_{j=1}^N \int_0^T (4h_j + \mathcal{O}(h^2)) |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \\
 &\quad - \frac{1}{12} \sum_{j=0}^N \int_0^T \left[\frac{2h}{g'_{j+1/2}} \left((\theta_0 + g_{j+1/2}) g''_{j+1/2} + (g'_{j+1/2})^2 \right) + \mathcal{O}(h^2) \right] |\psi'_{j+1} - \psi'_j|^2 dt \\
 &\quad - \alpha \sum_{j=1}^N \int_0^T h_j \phi'_j m_j dt \leq \theta_{N+1} \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + \frac{\theta_{N+1}}{6} \int_0^T |\psi'_N|^2 dt.
 \end{aligned} \tag{5.33}$$

There exists a constant $\hat{C}_{1,1} > 0$ independent of h such that

$$\frac{1}{12} \sum_{j=1}^N \int_0^T \mathcal{O}(h^2) |\psi'_j|^2 dt + \frac{1}{12} \sum_{j=0}^N \int_0^T \mathcal{O}(h^2) |\psi'_{j+1} - \psi'_j|^2 dt \leq \hat{C}_{1,1} h \mathcal{E}_h(0). \tag{5.34}$$

The application of the techniques used in the estimations of (4.36) and (5.18), combined with the estimate derived from Lemma 5.4, guarantees the existence of a constant $\hat{C}_{1,2} > 0$ such that the following holds:

$$\begin{aligned}
 &\alpha \sum_{j=1}^N \int_0^T \left(\frac{h_{j+1/2}}{6} \phi'_{j+1} + \frac{2h_j}{3} \phi'_j + \frac{h_{j-1/2}}{6} \phi'_{j-1} \right) m_j dt \\
 &= \alpha \left(\sum_{j=1}^N \int_0^T \frac{1}{h_j} \left(\frac{h_{j+1/2}}{6} \phi'_{j+1} + \frac{2h_j}{3} \phi'_j + \frac{h_{j-1/2}}{6} \phi'_{j-1} \right)^2 dt \right)^{\frac{1}{2}} \left(\sum_{j=1}^N \int_0^T h_j \left| \frac{m_j}{\theta_j} \right|^2 dt \right)^{1/2} \\
 &\leq \frac{\alpha(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T \frac{1}{h_j} \left(\frac{h_{j+1/2}}{6} \phi'_{j+1} + \frac{2h_j}{3} \phi'_j + \frac{h_{j-1/2}}{6} \phi'_{j-1} \right)^2 dt + \frac{\alpha(1 + \theta_0)}{2} \sum_{j=1}^N \int_0^T h_j \left| \frac{m_j}{\theta_j} \right|^2 dt \\
 &\leq \frac{\alpha(1 + \theta_0)}{12} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt + \frac{\alpha(1 + \theta_0)}{6} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt \\
 &\quad + \alpha(1 + \theta_0) \sum_{j=0}^N \int_0^T \left(2h_{j+1/2} + \mathcal{O}(h^2) \right) \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt + [1 + \theta_0 + \hat{C}_{1,2} h^2] \mathcal{E}_h(0)
 \end{aligned} \tag{5.35}$$

$$\begin{aligned} &\leq \frac{\alpha(1+\theta_0)}{12} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt + \frac{\alpha(1+\theta_0)}{6} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + [1+\theta_0 + \hat{C}_{1,2} h^2] \mathcal{E}_h(0) \\ &\quad + 2\alpha(1+\theta_0) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt + [1+\theta_0 + \hat{C}_{1,3} h^2] \mathcal{E}_h(0) \end{aligned}$$

The same arguments as in (5.35), together with the conservation energy property (5.7), yield the following result:

$$\begin{aligned} &\sum_{j=1}^N \left(\frac{h_{j+1/2}}{6} \psi'_{j+1} + \frac{2h_j}{3} \psi'_j + \frac{h_{j-1/2}}{6} \psi'_{j-1} \right) m_j \Big|_0^T \\ &\leq \frac{\alpha(1+\theta_0)}{12} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt + \frac{\alpha(1+\theta_0)}{6} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \hat{C}_{1,2} h^2 \mathcal{E}_h(0) \\ &\quad + \alpha(1+\theta_0) \sum_{j=0}^N \int_0^T \left(2h_{j+1/2} + \mathcal{O}(h^2) \right) \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ &\leq 2[4(1+\theta_0) + \hat{C}_{1,3} h^2] \mathcal{E}_h(0). \end{aligned} \tag{5.36}$$

In view of the inequalities (5.34), (5.35), and (5.36), the estimation in (5.33) yields:

$$\begin{aligned} &\left[1 - \frac{\alpha(\theta_0+1)}{2} \right] \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \right) \\ &\quad + [1 - 2\alpha(\theta_0+1)] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ &\quad - \frac{1}{12} \sum_{j=0}^N \int_0^T \frac{2h}{g'_{j+1/2}} \left((\theta_0 + g_{j+1/2}) g''_{j+1/2} + (g'_{j+1/2})^2 \right) |\psi'_{j+1} - \psi'_j|^2 dt \\ &\leq (\theta_0+1) \sum_{j=1}^N \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + \frac{\theta_0+1}{6} \int_0^T |\psi'_N|^2 dt \\ &\quad + [9(\theta_0+1) + \hat{C}_{1,1} h + 2\hat{C}_{1,2} h^2 + \hat{C}_{1,3} h^2] \mathcal{E}_h(0) \end{aligned} \tag{5.37}$$

The proof is then concluded by taking $\theta = \theta_0$ in the multiplier (4.28), and considering Eq. (4.38). $\square$

5.2. Proof of the observability inequality (3.2)

We now proceed to establish the estimate presented in Theorem 3.2. Combining the results of Lemma 5.3 and Lemma 5.4, we derive

$$\begin{aligned} &\frac{1}{2} \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt \right) \\ &\quad + \left(1 - \frac{\alpha\alpha_0}{2} \right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \\ &\quad - \frac{1}{2} (3 + \alpha) \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \right) \\ &\leq \left[2 \max(1, \alpha_0) + \frac{3}{\alpha} + \hat{C}_1 h^2 \right] \mathcal{E}_h(0). \end{aligned} \tag{5.38}$$

Now, by combining the above estimate with the one derived from Lemma 5.5, we get

$$\begin{aligned} &\frac{1}{2} [1 - \alpha(2\theta+3)] \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \right) \\ &\quad + 2[1 - 2\alpha(\theta+1)] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\ &\quad + \frac{1}{2} \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt \right) \\ &\quad + \left(1 - \frac{\alpha\alpha_0}{2} \right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt + \leq 2(\theta+1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt \\ &\quad + \frac{(\theta+1)}{3} \int_0^T |\psi'_N|^2 dt + [2C_1(\theta) + 2\hat{C}_2 h + \hat{C}_1 h^2 + 2 \max(1, \alpha_0) + \frac{3}{\alpha}] \mathcal{E}_h(0). \end{aligned} \tag{5.39}$$

Based on the energy conservation property (5.7) and the preceding estimate, there exists a positive constant C_2 that is independent of h .

$$\frac{1}{2} [1 - 3\alpha] \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\psi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\psi'_{j+1} + \psi'_j|^2 dt \right)$$

$$\begin{aligned}
& + 2[1 - 2\alpha] \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\psi_{j+1} - \psi_j}{h_{j+1/2}} \right|^2 dt \\
& + \frac{1}{2} \left(\frac{1}{3} \sum_{j=1}^N \int_0^T h_j |\varphi'_j|^2 dt + \frac{1}{6} \sum_{j=0}^N \int_0^T h_{j+1/2} |\varphi'_{j+1} + \varphi'_j|^2 dt \right) \\
& + \left(1 - \frac{\alpha\alpha_0}{2} \right) \sum_{j=0}^N \int_0^T h_{j+1/2} \left| \frac{\varphi_{j+1} - \varphi_j}{h_{j+1/2}} \right|^2 dt \leq 2(\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt \\
& + \frac{(\theta + 1)}{3} \int_0^T |\psi'_N|^2 dt + [2C_1(\theta) + 8T\theta\alpha + C_2h + 2\max(1, \alpha_0) + \frac{3}{\alpha}] \mathcal{E}_h(0).
\end{aligned} \tag{5.40}$$

Let α lie within the interval $(0, \hat{\alpha})$, and let $M(\alpha)$ be defined as in (4.44). Denoting $\hat{C}$ as the constant multiplier in the energy in (5.40), the following inequality holds

$$TM(\alpha)\mathcal{E}_h(0) - \hat{C}(\alpha, h, \theta)\mathcal{E}_h(0) \leq 2(\theta + 1) \int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + \frac{(\theta + 1)}{3} \int_0^T |\psi'_N|^2 dt. \tag{5.41}$$

The argument to complete the proof follows similarly to the finite difference framework. For this, let $T_0(r)$, $r > 0$ be the threshold defined in (4.46), and let ϵ in $(0, 1)$, such that the following two inequalities are satisfied:

$$8\alpha\theta \leq \epsilon M(\alpha), \quad 2C_1(\theta) + 2C_2h + 2\max(1, \alpha_0) + \frac{3}{\alpha} \leq (1 - \epsilon)M(\alpha)T_0(r). \tag{5.42}$$

In light of (5.41) and (5.42), for a given time T that exceeds the threshold $T_0(r)$, the following inequality holds:

$$\mathcal{E}_h(0) \leq C_r(\alpha, T, \theta) \left[\int_0^T \left| \frac{\psi_N}{h_{N+1/2}} \right|^2 dt + \frac{1}{6} \int_0^T |\psi'_N|^2 dt \right],$$

where

$$C_r(\alpha, T, \theta) := \frac{2(\theta + 1)}{(1 - \epsilon)M(\alpha)(T - T_0(r))}.$$

Thus, the proof of Theorem 3.2 is complete if the second equation in (5.42) is satisfied. This condition holds for $r > 0$ sufficiently large and h sufficiently small. Furthermore, the constant in the estimate (3.2) and the threshold time T_0 in Theorem 3.2 are determined by selecting the pair (r, θ) such that r is as small as possible while remaining within the largest possible interval for θ values.

Remark 5.6.

- The observation point addressed in Remark 4.8, within the context of the finite difference method, is similarly valid in the context of the finite element method.
- The minimal $T_0(r)$ for the observability inequality (3.2) remains unchanged from that in the finite difference case. A similar observation holds for the parameter θ .
- The interval $(0, \hat{\alpha})$ of values for α for which the observability inequality holds uniformly with respect to h , whether in the finite difference or finite element setting, is the same as that provided in the continuous setting.³

6. Comments and perspectives

- This work successfully generalizes discretization concepts on uniform meshes via convex mapping, achieving uniform observability inequalities (3.1) and (3.2) as a first approach in the context of strongly coupled waves. However, several limitations emerge: First, the discrete observability problem is solved only for threshold times T_0 larger than the continuous threshold T_0 , which remains an open problem in this context.

Secondly, a constraint on the selection of θ exists, in contrast to the flexibility observed in single wave equation. It remains an open question whether this limitation is inherent when the coupling parameter α is small, or if a greater range of possible choices for θ could be achieved, as in the single wave context.

- As in the continuous setting, when employing multiplier techniques, the observability inequality is obtained only for restricted intervals of the coupling parameter. For different algebraic properties of α the observability inequality has been recovered in Ref. 3 using Ingham-type theorems.³¹ Extending this approach to the discrete level on non-uniform meshes would first require an empirical analysis of the eigenvalue gap uniformity as h approaches zero, followed by theoretical estimation of these gaps in terms of the eigenvalues of the single wave equation.
- In Ref. 9, the authors present a finite-dimensional framework demonstrating the equivalence between the uniformly bounded control property of a controlled system and the uniform observability inequality of the associated adjoint system as $h \rightarrow 0$. Consequently, the uniform observability inequalities (3.1) and (3.2) may lead to uniformly bounded discrete control v_h for the discretization of the following controlled system:

$$\begin{cases} q_{tt} - q_{xx} + \alpha y_t = 0, & (x, t) \in (0, 1) \times (0, T), \\ y_{tt} - \psi_{xx} - \alpha q_t = 0, & (x, t) \in (0, 1) \times (0, T), \\ q(0, t) = q(1, t) = 0, & t \in (0, T), \\ y(0, t) = 0, \quad y(1, t) = v(t), & (0, T), \\ q(x, T) = q^0, \quad q_t(x, T) = q^1(x), & x \in (0, 1), \\ y(x, T) = y^0(x), \quad y_t(x, T) = y^1(x), & x \in (0, 1), \end{cases} \tag{6.1}$$

These discretizations can be implemented using finite difference and finite element methods on non-uniform meshes. The numerical implementation of the control computation is based on the algorithm outlined in Ref. 10, which relies on the Hilbert Uniqueness Method (HUM). This approach requires time discretization of the system (6.1), raising the important question of stability. To accurately capture the physical behavior of system (6.1), it is crucial to carefully consider the interaction between the time step Δt , the spatial discretization parameter h , and the coefficient α .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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